

GWANYEOP KIM

Undergraduate Student

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📍 Konkuk University, Seoul

RESEARCH INTERESTS

My research interests lie in building reliable and robust agentic AI systems capable of reasoning, collaboration, and multimodal interaction. I am particularly interested in LLM-based multi-agent systems, multimodal agentic AI, retrieval-augmented generation, and methods for improving the reliability and robustness of AI agents, ultimately toward the development of generalist AI agents.

EDUCATION

Konkuk University | Mar.2025 - Present

Dept. Computer Science & Engineering

Republic of Korea Air Force | Mar.2023 - Dec.2024

Military Service

Information System Management Specialist

Sejong University | Mar.2021 - Dec.2022

Dept. Computer Science & Engineering

RESEARCH EXPERIENCE

Natural Language Processing Lab (NLP Lab), UNIST | May.2025 - Present

Undergraduate Research Intern

- Research Area: RAG Optimization, Efficient Reasoning, Agentic LLM Systems
- Conducted research on improving RAG systems through adaptive optimization strategies
- Designed and implemented reward-based retrieval-generation balancing mechanisms
- Led the development of a research prototype resulting in a first-author paper under review

Data Mining and Information Systems Lab (DMIS Lab), Korea University | Mar.2026 - Jul.2026

Undergraduate Research Intern

- Research Area: Agentic AI, Multi Agent System, BioNLP

PUBLICATIONS

Unpublished Preprint

"AMORO-RAG : Adaptive Multi-Objective Reinforcement Optimization for RAG"

- 1st author
- Proposed a dynamic RAG optimization framework that improves retrieval quality and reasoning reliability through query-specific reward weighting
- Designed query-specific reward weighting mechanism

AWARDS

Hosted/Organized by : Ministry of Trade, Industry and Energy / Korea Robot Industry Advancement Institute, Korea Robot Industry Association

2021 16th IRC AI Robot Autonomous Mission Competition

- in the Autonomous Flight Category
 - Gold Prize (1st Place)
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SKILLS AND CERTIFICATIONS

- Language : English (Intermediate), Korean (Native)
 - Programming Languages : Python (Advanced), C (Intermediate)
 - Data Science & Machine Learning : Deep Learning/NLP (Advanced), Machine Learning (Advanced), Computer Vision (Basic)
 - Frameworks and Libraries : PyTorch/HuggingFace/FAISS (Advanced), TensorFlow/LangChain/NumPy/Pandas (Intermediate)
 - Certificates : SQLD, Information Processing Technician
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PROJECTS

Konkuk University CSE graduation project | Feb.2026 - Dec.2026

Agentic RAG-based Medication Recommendation System

- Developed an agentic RAG-based system for symptom-driven medication recommendation, where the model performs iterative retrieval and analysis to infer appropriate over-the-counter (OTC) drugs.
- Designed a structured decision pipeline that constrains recommendations to safe, non-prescription medications, improving reliability and safety in domain-specific generation.

Konkuk University KES Academic Seminar | Jul.2026 - Aug.2026

Multi-modal AI-based road and pedestrian infrastructure civil complaint analysis and Hazardous Area Maintenance Decision Support System

- Developed a multi-modal AI system that integrates YOLO-based visual damage detection and LLM-based civil complaint analysis to identify road and pedestrian infrastructure hazards.
 - Designed a rule- and RAG-grounded multi-modal decision pipeline that combines visual and textual evidence to infer damage severity, urgency, responsible departments, and maintenance actions.
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ACTIVITIES

- LG Aimers 7th Cohort | Jul.2025 - Aug.2025
- 2025 Dacon Food and Beverage Menu Demand Prediction AI Online Hackathon, Top 10%